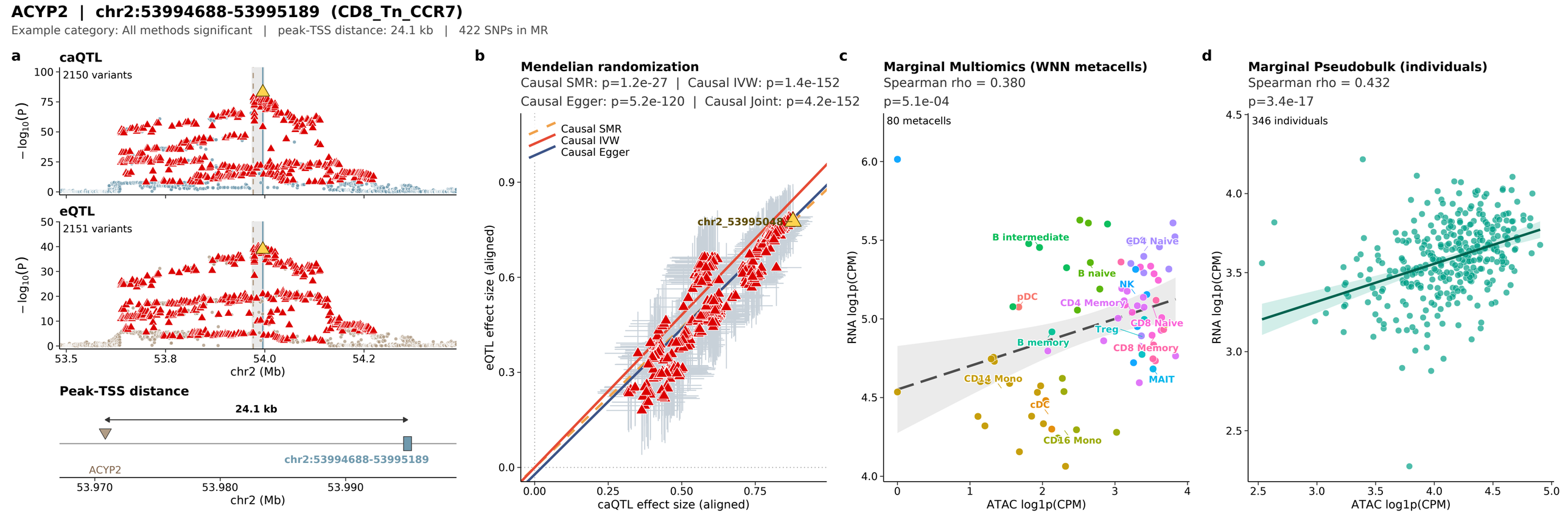
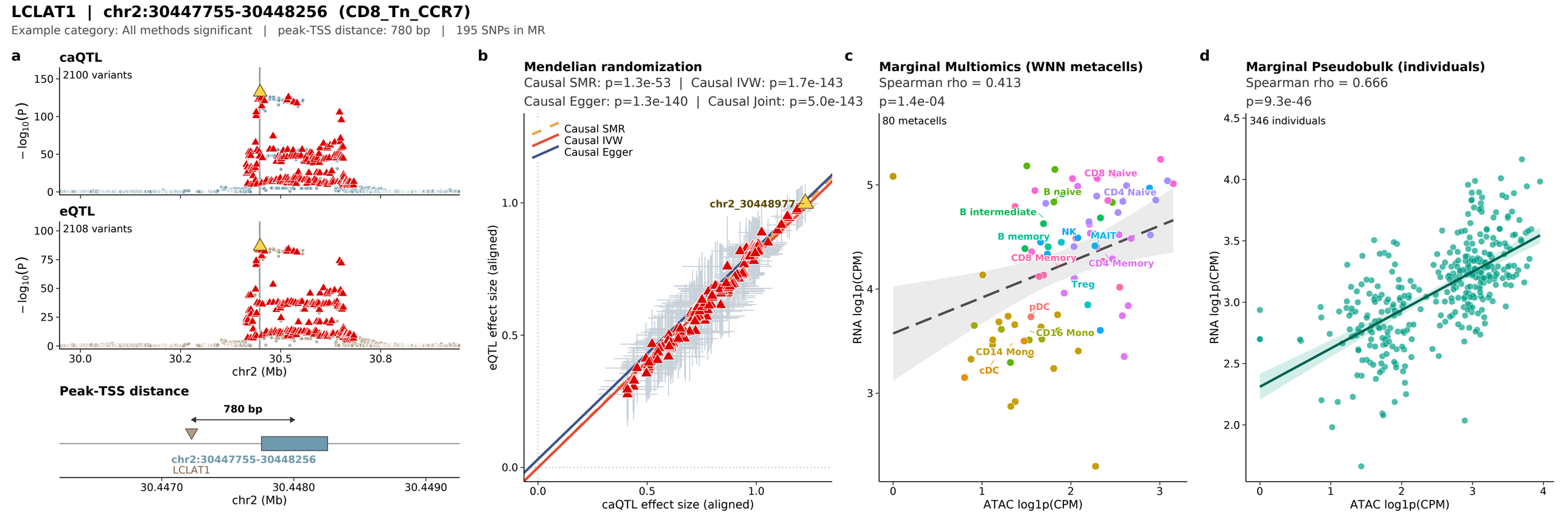
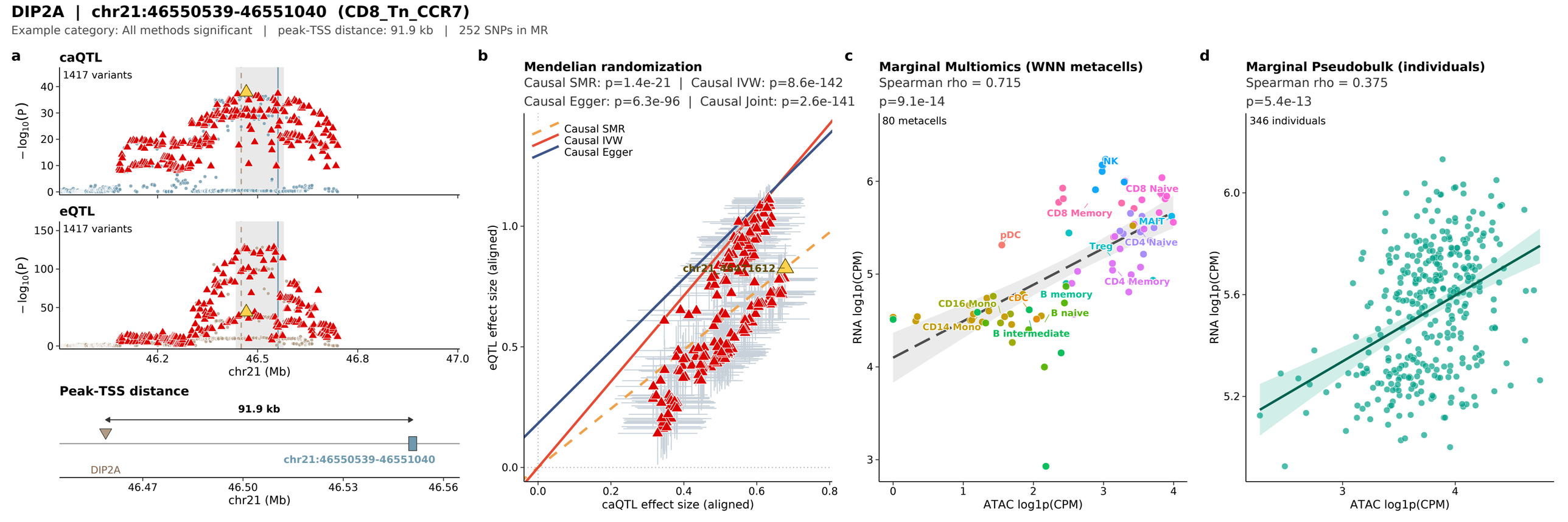
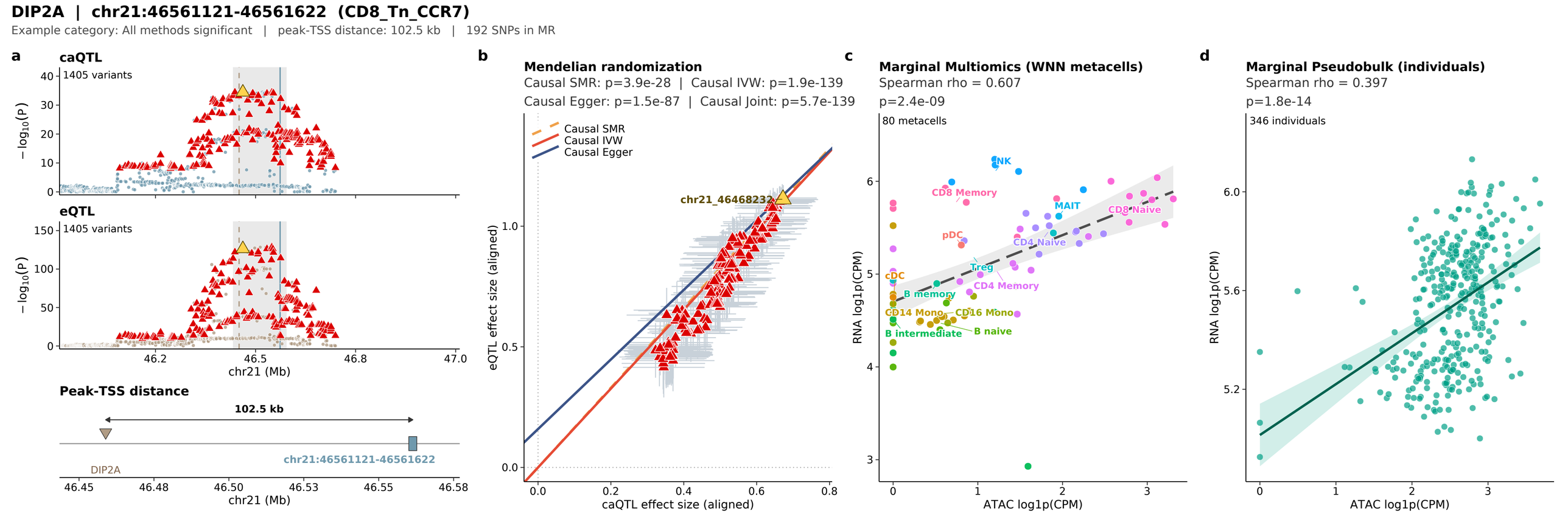

ACYP2 | chr2:53994688-53995189 (CD8_Tn_CCR7)
Example category: All methods significant | peak-TSS distance: 24.1 kb | 422 SNPs in MR
a
caQTL
2150 variants
eQTL
2151 variants
− log10(P)
chr2 (Mb)
Peak-TSS distance
24.1 kb
chr2:53994688-53995189
ACYP2
b
Mendelian randomization
Causal SMR: p=1.2e-27 | Causal IVW: p=1.4e-152
Causal Egger: p=5.2e-120 | Causal Joint: p=4.2e-152
Causal SMR
Causal IVW
Causal Egger
chr2_53995049
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.380
p=5.1e-04
80 metacells
B intermediate
CD4 Naive
B naive
NK
pDC
CD4 Memory
CD8 Naive
Treg
B memory
CD8 Memory
MAIT
CD14 Mono
cDC
CD16 Mono
RNA log1p(CPM)
ATAC log1p(CPM)
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.432
p=3.4e-17
346 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

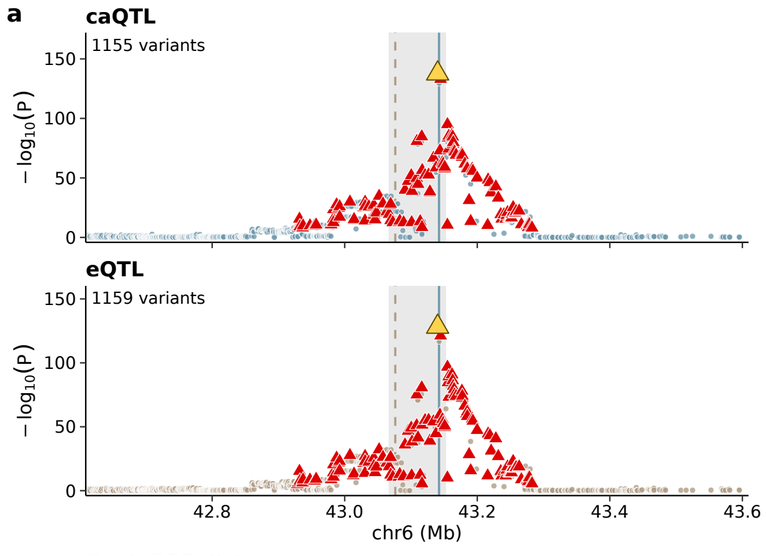

LCLAT1 | chr2:30447755-30448256 (CD8_Tn_CCR7)
Example category: All methods significant | peak-TSS distance: 780 bp | 195 SNPs in MR
a
caQTL
2100 variants
eQTL
2108 variants
− log10(P)
chr2 (Mb)
Peak-TSS distance
780 bp
chr2:30447755-30448256
LCLAT1
b
Mendelian randomization
Causal SMR: p=1.3e-53 | Causal IVW: p=1.7e-143
Causal Egger: p=1.3e-140 | Causal Joint: p=5.0e-143
Causal SMR
Causal IVW
Causal Egger
chr2_30448977
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.413
p=1.4e-04
80 metacells
CD8 Naive
B naive
CD4 Naive
B intermediate
B memory
NK
MAIT
CD8 Memory
CD4 Memory
Treg
pDC
CD16 Mono
CD14 Mono
cDC
RNA log1p(CPM)
ATAC log1p(CPM)
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.666
p=9.3e-46
346 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

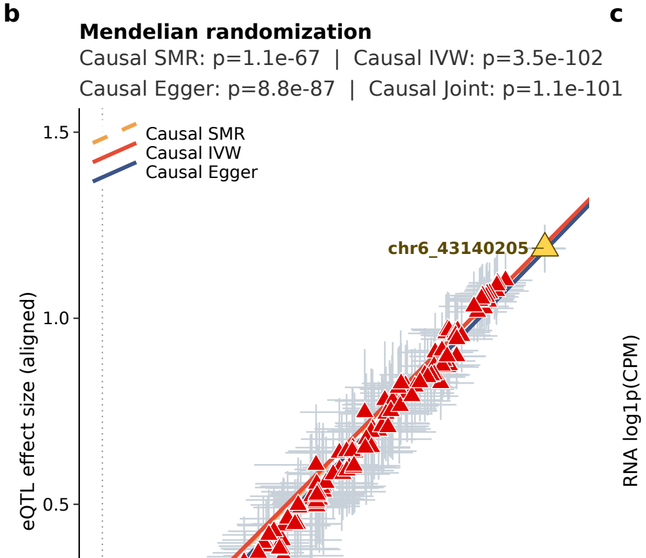

DIP2A | chr21:46550539-46551040 (CD8_Tn_CCR7)
Example category: All methods significant | peak-TSS distance: 91.9 kb | 252 SNPs in MR
a
caQTL
1417 variants
eQTL
1417 variants
− log10(P)
chr21 (Mb)
Peak-TSS distance
91.9 kb
chr21:46550539-46551040
DIP2A
b
Mendelian randomization
Causal SMR: p=1.4e-21 | Causal IVW: p=8.6e-142
Causal Egger: p=6.3e-96 | Causal Joint: p=2.6e-141
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.715
p=9.1e-14
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
NK
CD8 Naive
CD8 Memory
MAIT
pDC
Treg
CD4 Naive
CD4 Memory
CD16 Mono
cDC
B memory
B naive
CD14 Mono
B intermediate
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.375
p=5.4e-13
346 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

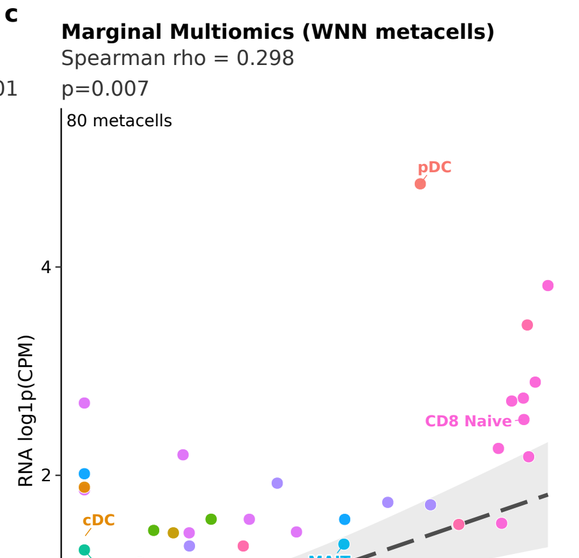

DIP2A | chr21:46561121-46561622 (CD8_Tn_CCR7)
Example category: All methods significant | peak-TSS distance: 102.5 kb | 192 SNPs in MR
a
caQTL
1405 variants
eQTL
1405 variants
chr21 (Mb)
Peak-TSS distance
102.5 kb
chr21:46561121-46561622
DIP2A
b
Mendelian randomization
Causal SMR: p=3.9e-28 | Causal IVW: p=1.9e-139
Causal Egger: p=1.5e-87 | Causal Joint: p=5.7e-139
Causal SMR
Causal IVW
Causal Egger
chr21_46468232
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.607
p=2.4e-09
80 metacells
NK
CD8 Memory
MAIT
CD8 Naive
pDC
CD4 Naive
Treg
cDC
CD4 Memory
B memory
CD14 Mono
CD16 Mono
B intermediate
B naive
RNA log1p(CPM)
ATAC log1p(CPM)
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.397
p=1.8e-14
346 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

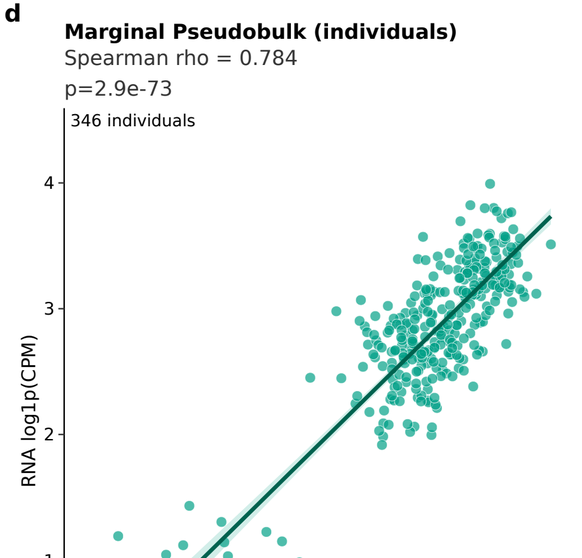
PTK7 | chr6:43142078-43142579 (CD8_Tn_CCR7)

Example category: All methods significant | peak-TSS distance: 66.0 kb | 127 SNPs in MR



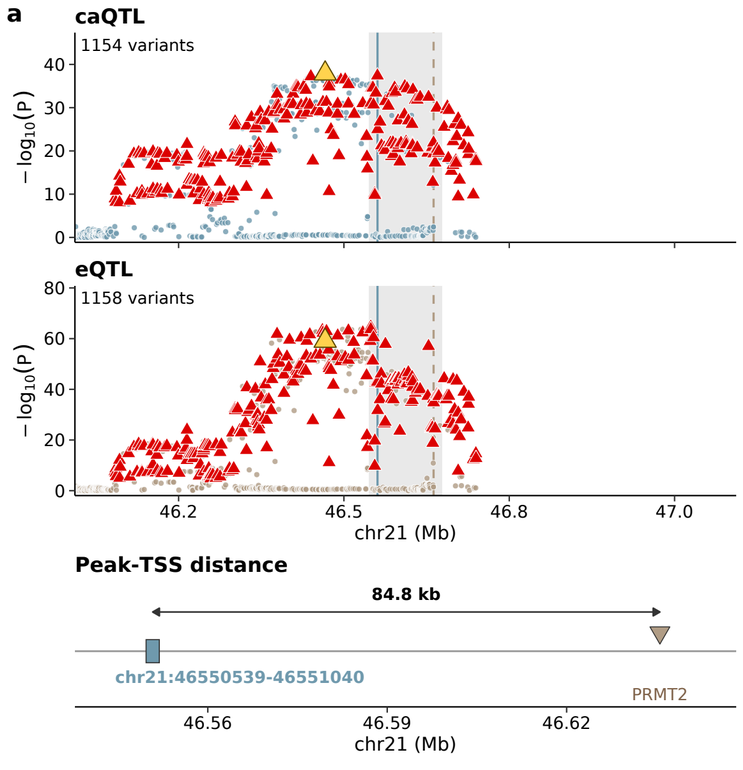


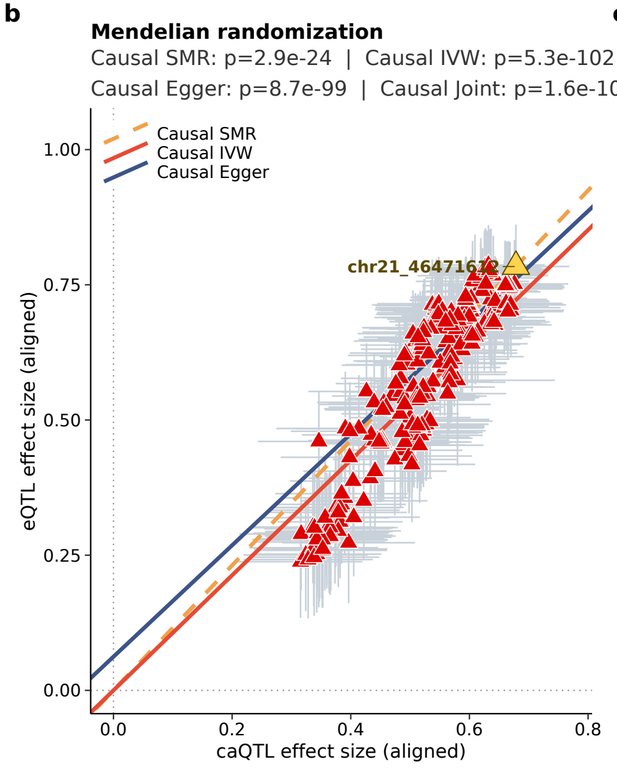


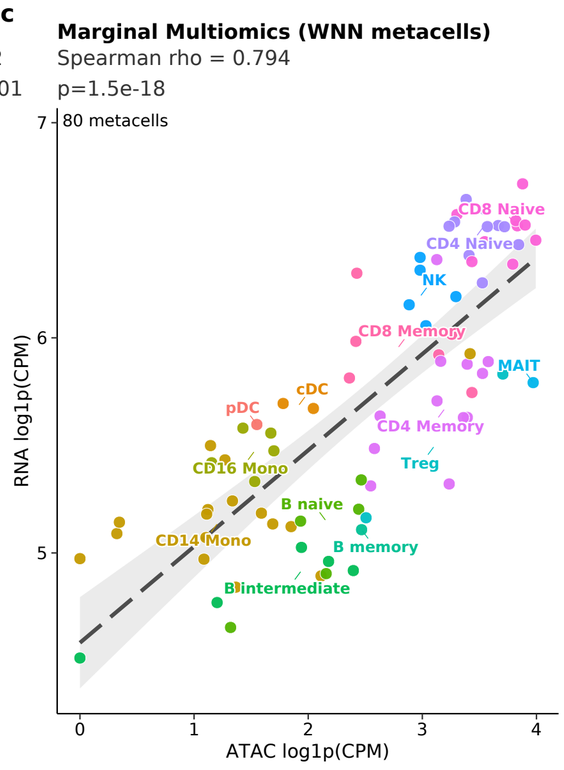


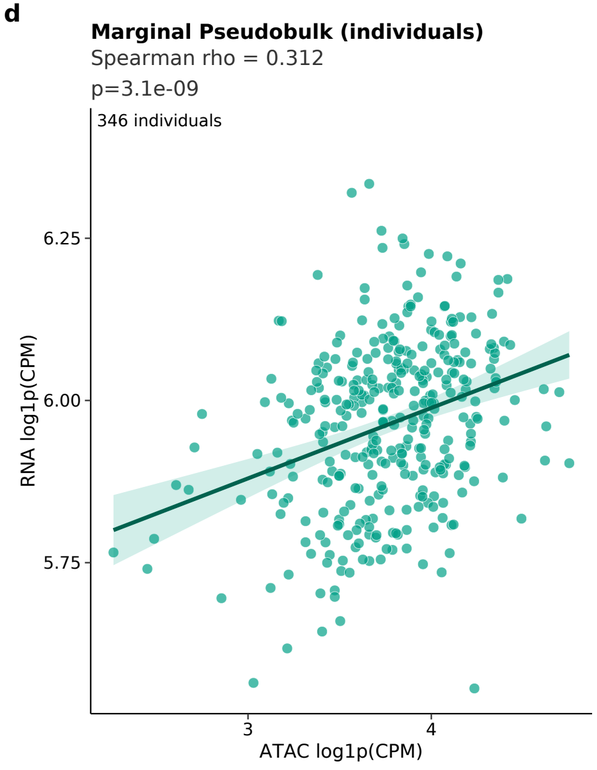
PRMT2 | chr21:46550539-46551040 (CD8_Tn_CCR7)

Example category: All methods significant | peak-TSS distance: 84.8 kb | 249 SNPs in MR



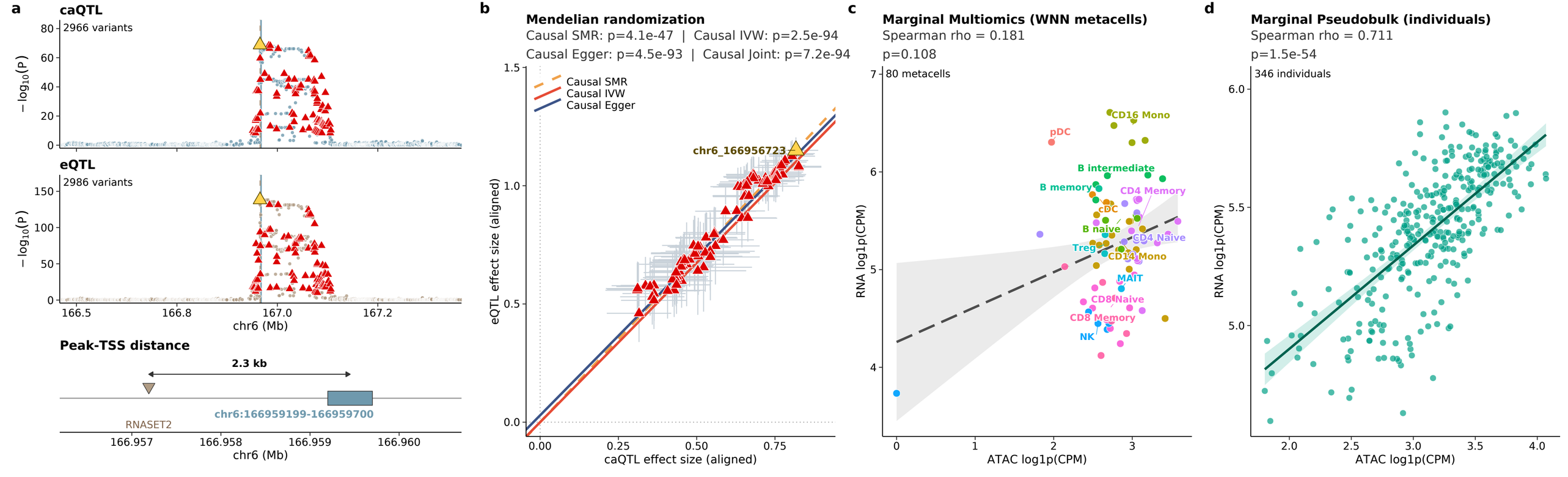


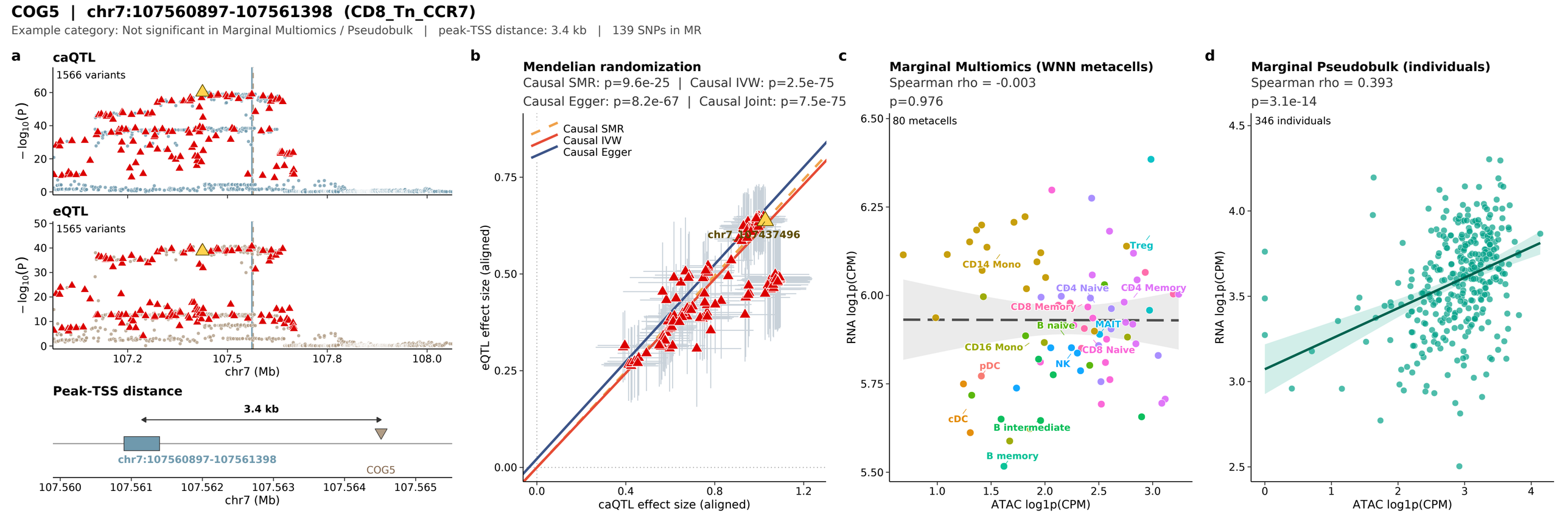


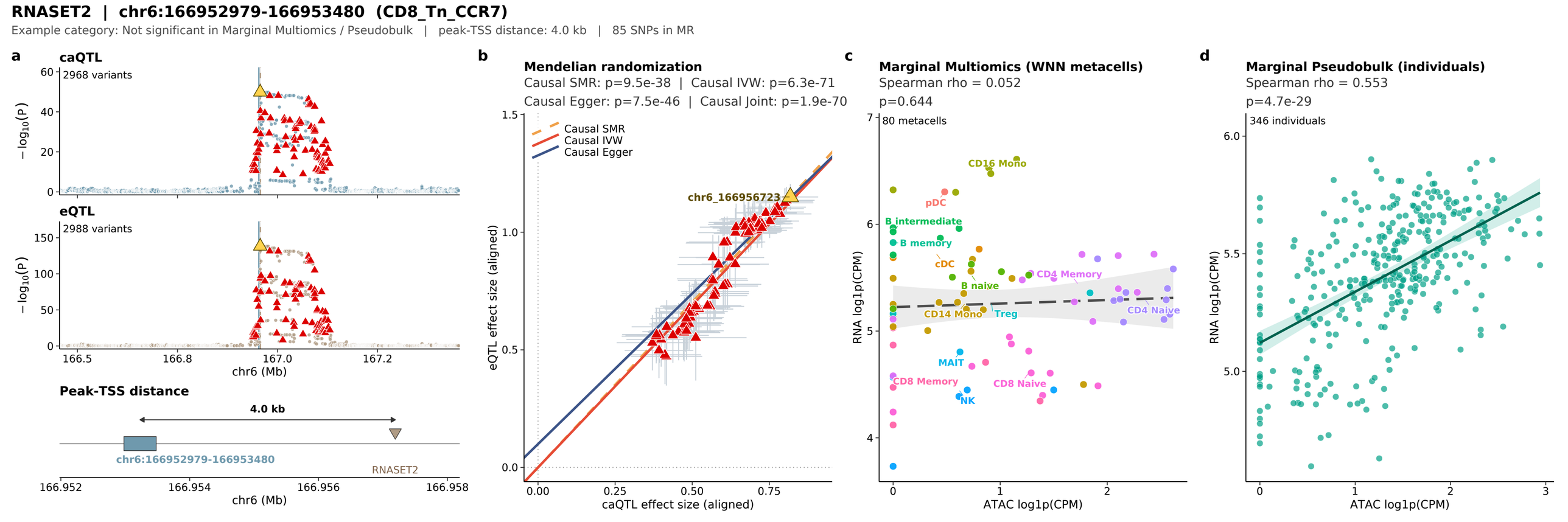


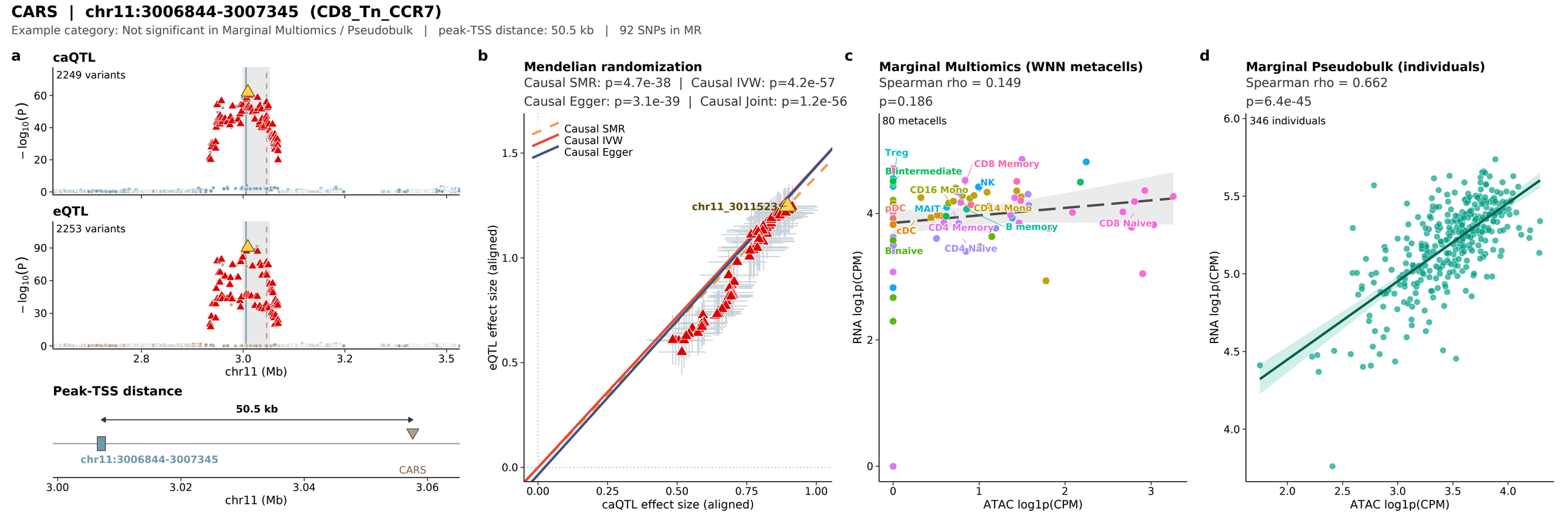
RNASET2 | chr6:166959199-166959700 (CD8_Tn_CCR7)

Example category: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 2.3 kb | 91 SNPs in MR



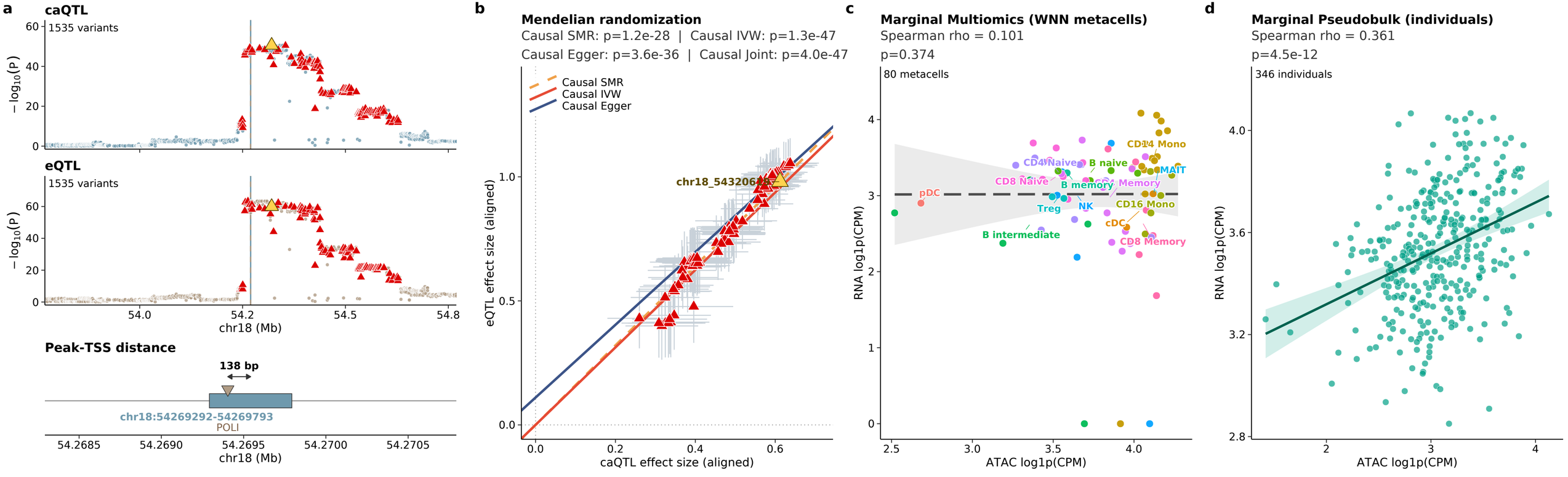

COG5 | chr7:107560897-107561398 (CD8_Tn_CCR7)
Example category: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 3.4 kb | 139 SNPs in MR
a
caQTL
1566 variants
eQTL
1565 variants
− log10(P)
chr7 (Mb)
Peak-TSS distance
3.4 kb
chr7:107560897-107561398
COG5
b
Mendelian randomization
Causal SMR: p=9.6e-25 | Causal IVW: p=2.5e-75
Causal Egger: p=8.2e-67 | Causal Joint: p=7.5e-75
Causal SMR
Causal IVW
Causal Egger
chr7:107437496
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = -0.003
p=0.976
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
CD14 Mono
Treg
CD4 Naive
CD4 Memory
CD8 Memory
B naive
MAIT
CD16 Mono
CD8 Naive
pDC
NK
cDC
B intermediate
B memory
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.393
p=3.1e-14
346 individuals
RNA log1p(CPM)
ATAC log1p(CPM)


RNASET2 | chr6:166952979-166953480 (CD8_Tn_CCR7)
Example category: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 4.0 kb | 85 SNPs in MR
a
caQTL
2968 variants
eQTL
2988 variants
− log10(P)
chr6 (Mb)
Peak-TSS distance
4.0 kb
chr6:166952979-166953480
RNASET2
b
Mendelian randomization
Causal SMR: p=9.5e-38 | Causal IVW: p=6.3e-71
Causal Egger: p=7.5e-46 | Causal Joint: p=1.9e-70
Causal SMR
Causal IVW
Causal Egger
chr6_166956723
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.052
p=0.644
80 metacells
CD16 Mono
pDC
B intermediate
B memory
cDC
B naive
CD4 Memory
CD14 Mono
Treg
CD4 Naive
MAIT
CD8 Memory
CD8 Naive
NK
RNA log1p(CPM)
ATAC log1p(CPM)
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.553
p=4.7e-29
346 individuals
RNA log1p(CPM)
ATAC log1p(CPM)


CARS | chr11:3006844-3007345 (CD8_Tn_CCR7)
Example category: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 50.5 kb | 92 SNPs in MR
a
caQTL
2249 variants
eQTL
2253 variants
− log10(P)
chr11 (Mb)
Peak-TSS distance
50.5 kb
chr11:3006844-3007345
CARS
b
Mendelian randomization
Causal SMR: p=4.7e-38 | Causal IVW: p=4.2e-57
Causal Egger: p=3.1e-39 | Causal Joint: p=1.2e-56
Causal SMR
Causal IVW
Causal Egger
chr11_3011523
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.149
p=0.186
80 metacells
Treg
B intermediate
CD8 Memory
NK
CD16 Mono
pDC
MAIT
CD14 Mono
cDC
CD4 Memory
B memory
CD8 Naive
B naive
CD4 Naive
RNA log1p(CPM)
ATAC log1p(CPM)
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.662
p=6.4e-45
346 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

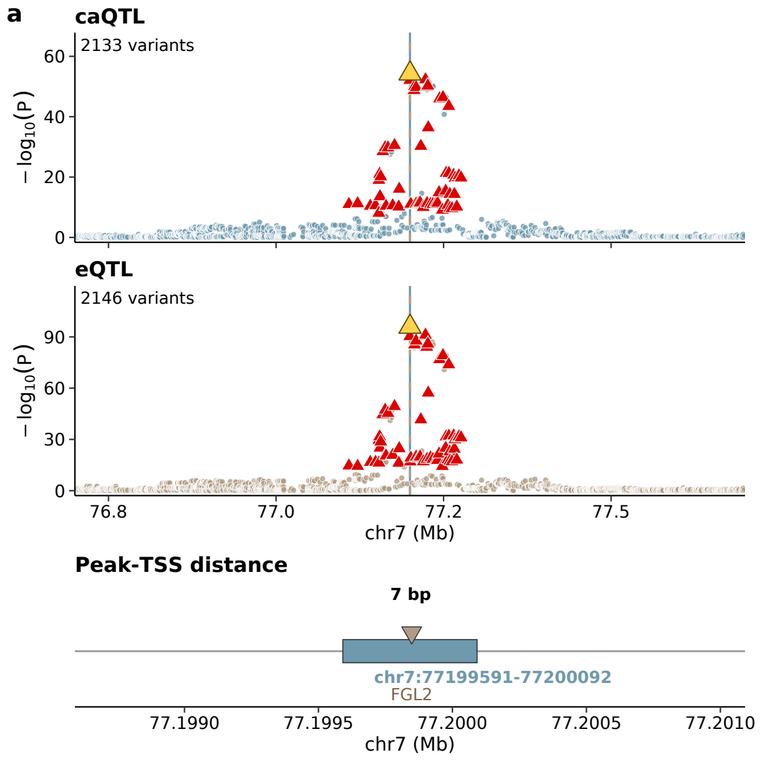
POLI | chr18:54269292-54269793 (CD8_Tn_CCR7)

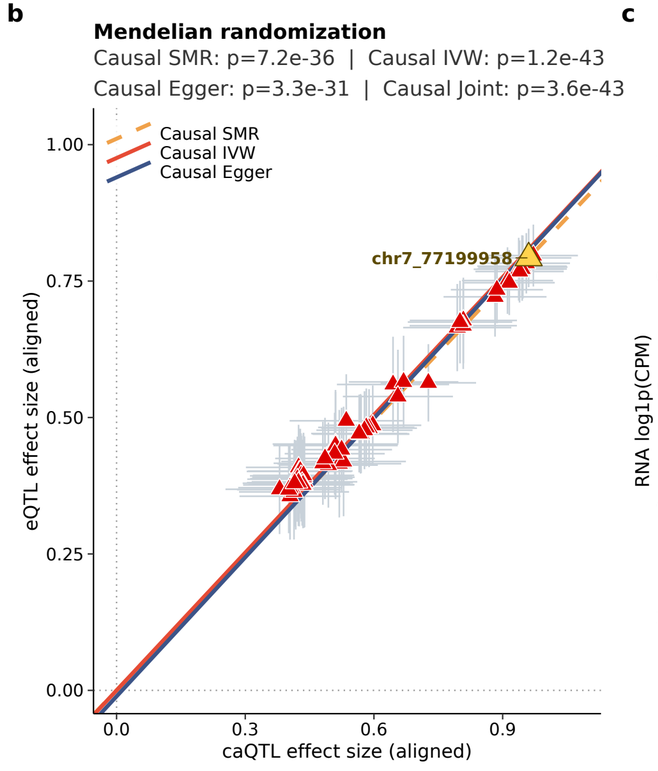
Example category: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 138 bp | 107 SNPs in MR

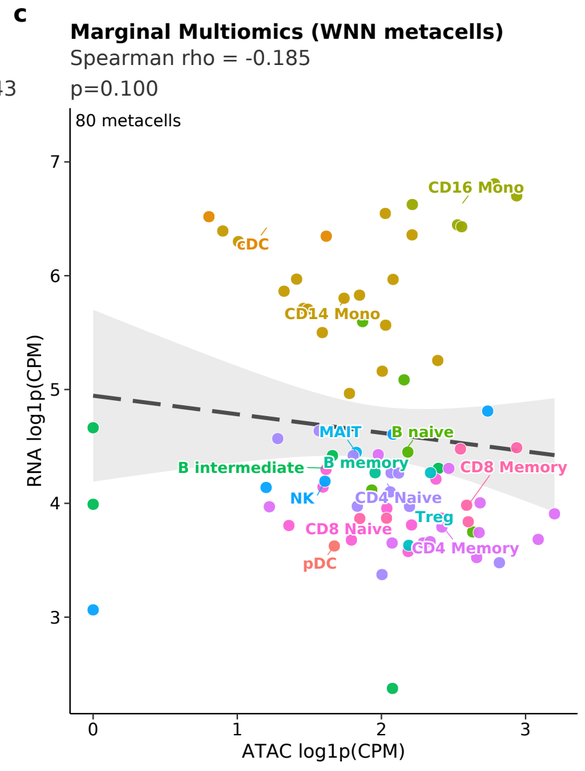


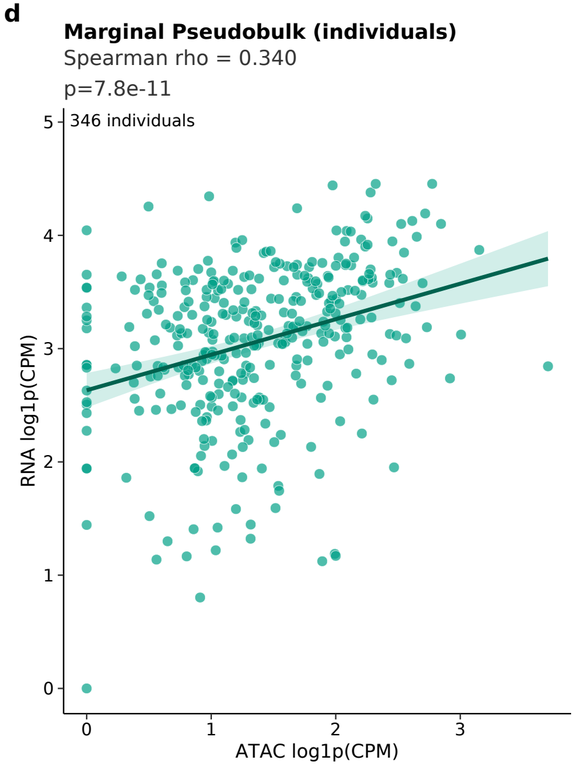
FGL2 | chr7:77199591-77200092 (CD8_Tn_CCR7)

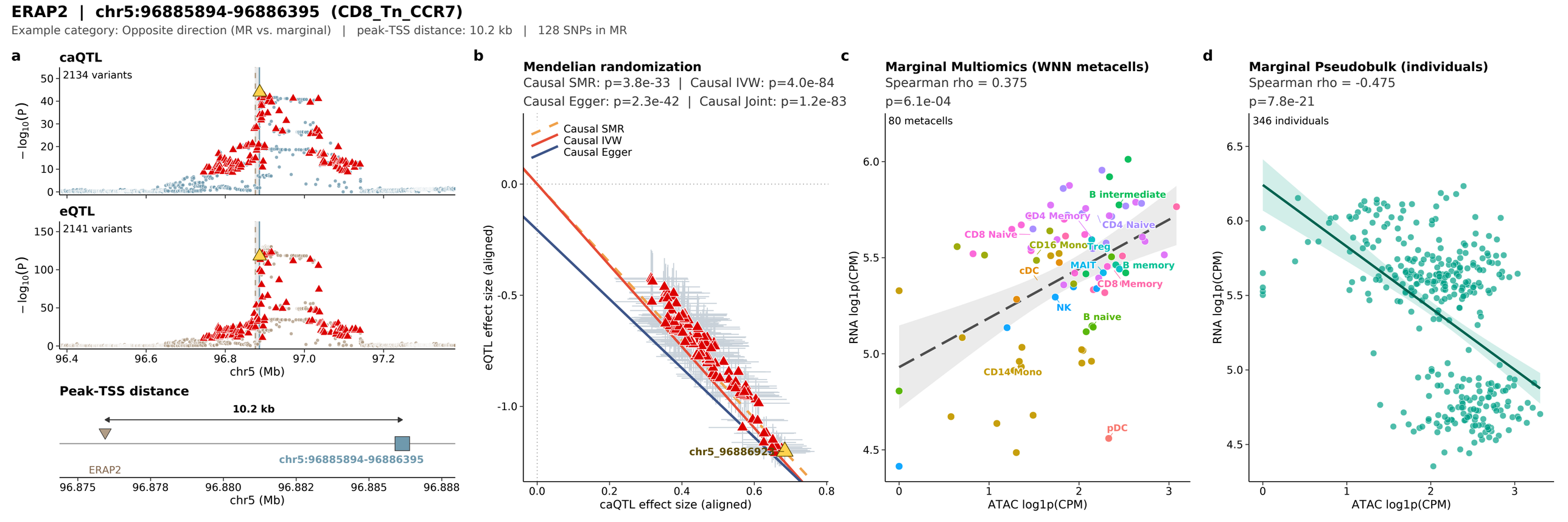
Example category: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 7 bp | 58 SNPs in MR

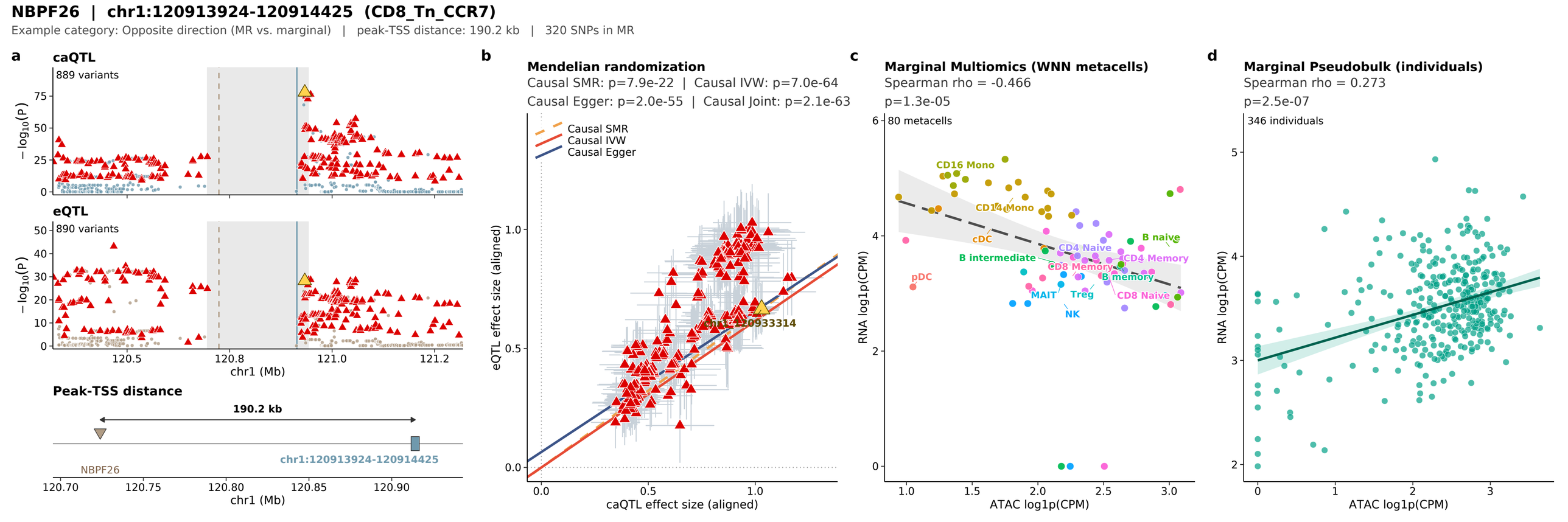


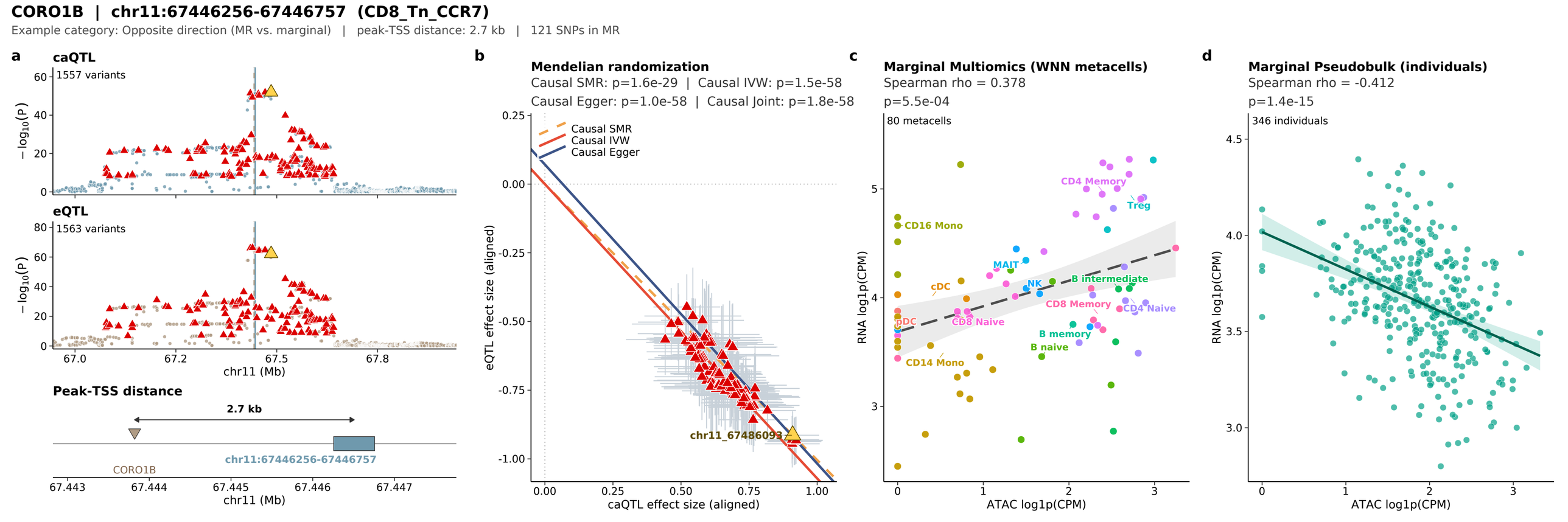


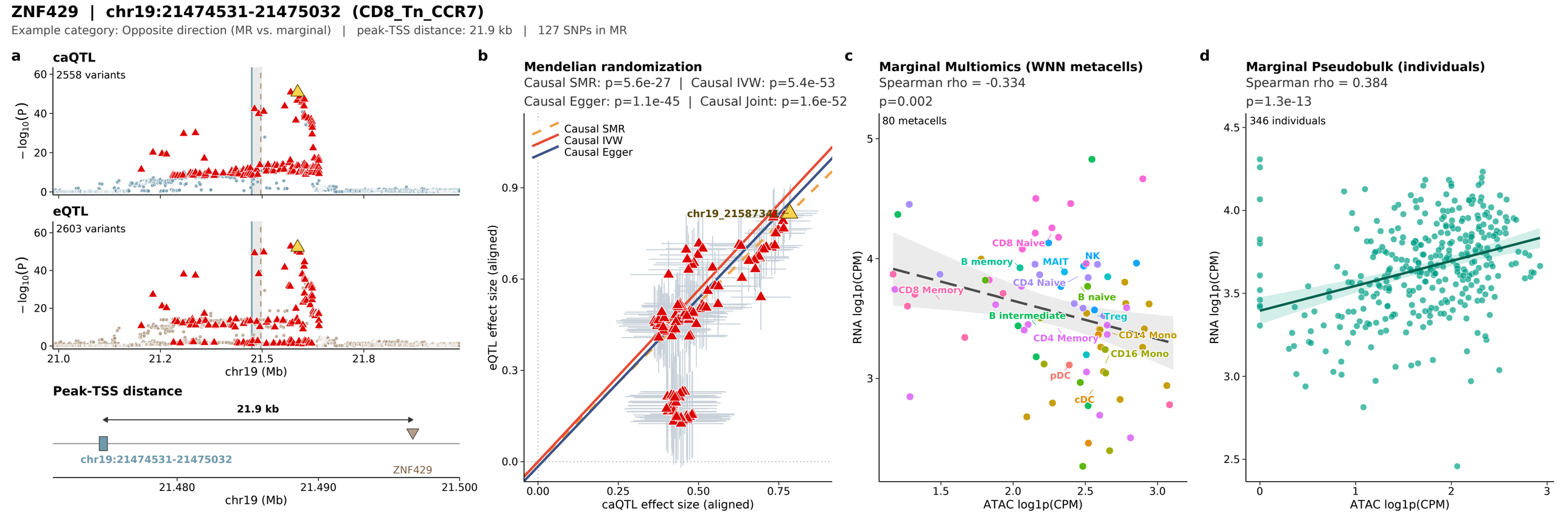


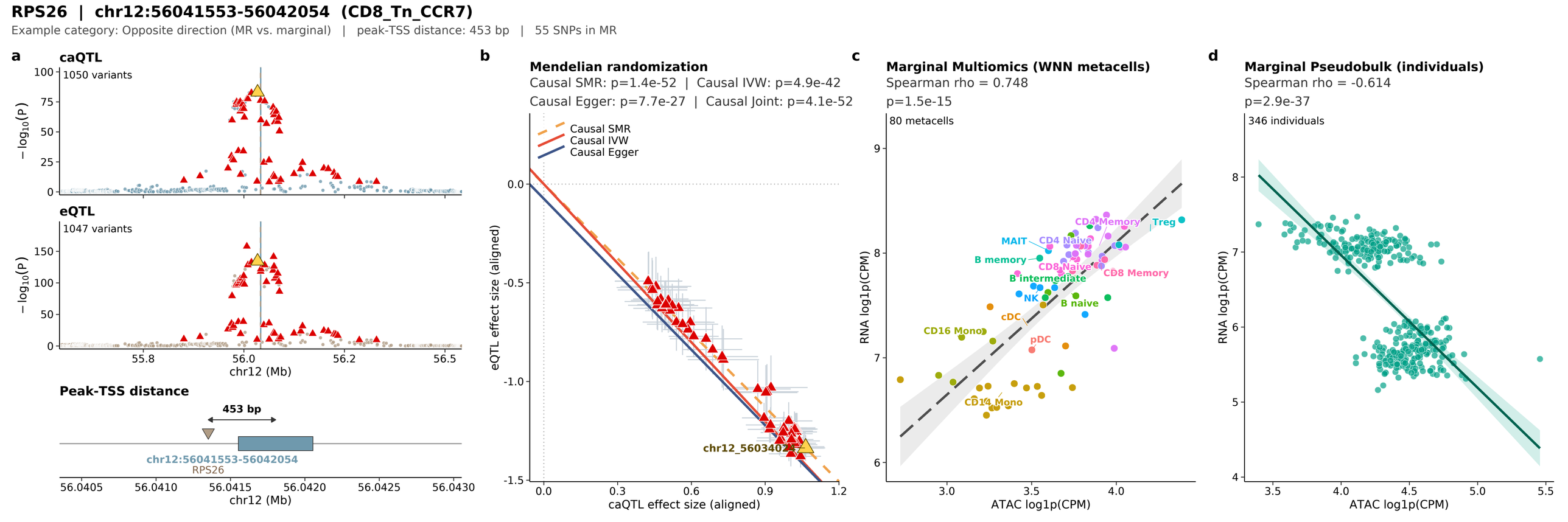



ERAP2 | chr5:96885894-96886395 (CD8_Tn_CCR7)
Example category: Opposite direction (MR vs. marginal) | peak-TSS distance: 10.2 kb | 128 SNPs in MR
a
caQTL
2134 variants
− log10(P)
eQTL
2141 variants
chr5 (Mb)
Peak-TSS distance
10.2 kb
ERAP2
chr5:96885894-96886395
b
Mendelian randomization
Causal SMR: p=3.8e-33 | Causal IVW: p=4.0e-84
Causal Egger: p=2.3e-42 | Causal Joint: p=1.2e-83
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.375
p=6.1e-04
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
B intermediate
CD4 Memory
CD4 Naive
CD8 Naive
CD16 Mono
Treg
MAIT
B memory
cDC
CD8 Memory
NK
B naive
CD14 Mono
pDC
d
Marginal Pseudobulk (individuals)
Spearman rho = -0.475
p=7.8e-21
346 individuals
RNA log1p(CPM)
ATAC log1p(CPM)


NBPF26 | chr1:120913924-120914425 (CD8_Tn_CCR7)
Example category: Opposite direction (MR vs. marginal) | peak-TSS distance: 190.2 kb | 320 SNPs in MR
a
caQTL
889 variants
eQTL
890 variants
Peak-TSS distance
190.2 kb
NBPF26
chr1:120913924-120914425
chr1 (Mb)
b
Mendelian randomization
Causal SMR: p=7.9e-22 | Causal IVW: p=7.0e-64
Causal Egger: p=2.0e-55 | Causal Joint: p=2.1e-63
Causal SMR
Causal IVW
Causal Egger
chr1:120933314
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = -0.466
p=1.3e-05
80 metacells
CD16 Mono
CD14 Mono
cDC
B intermediate
pDC
CD4 Naive
CD8 Memory
B naive
CD4 Memory
B memory
MAIT
Treg
NK
CD8 Naive
RNA log1p(CPM)
ATAC log1p(CPM)
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.273
p=2.5e-07
346 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

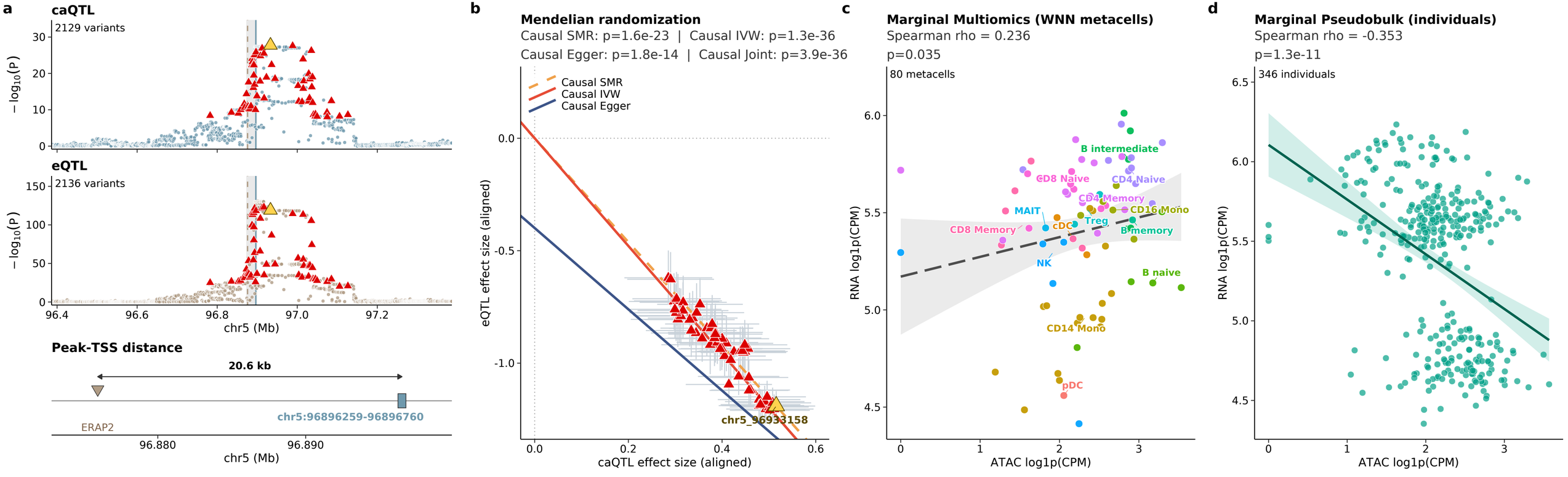

CORO1B | chr11:67446256-67446757 (CD8_Tn_CCR7)
Example category: Opposite direction (MR vs. marginal) | peak-TSS distance: 2.7 kb | 121 SNPs in MR
a
caQTL
1557 variants
eQTL
1563 variants
− log10(P)
chr11 (Mb)
Peak-TSS distance
2.7 kb
chr11:67446256-67446757
CORO1B
b
Mendelian randomization
Causal SMR: p=1.6e-29 | Causal IVW: p=1.5e-58
Causal Egger: p=1.0e-58 | Causal Joint: p=1.8e-58
Causal SMR
Causal IVW
Causal Egger
chr11_67486093
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.378
p=5.5e-04
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
CD4 Memory
Treg
CD16 Mono
MAIT
cDC
NK
B intermediate
CD8 Memory
CD4 Naive
pDC
CD8 Naive
B memory
B naive
CD14 Mono
d
Marginal Pseudobulk (individuals)
Spearman rho = -0.412
p=1.4e-15
346 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

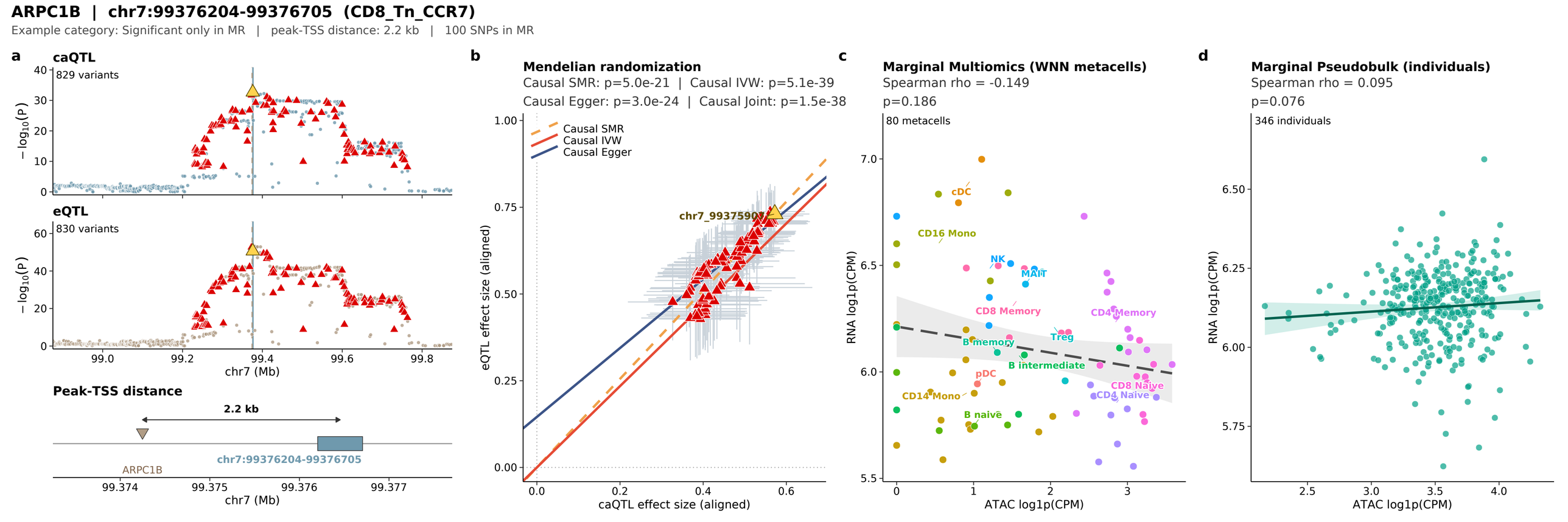

ZNF429 | chr19:21474531-21475032 (CD8_Tn_CCR7)
Example category: Opposite direction (MR vs. marginal) | peak-TSS distance: 21.9 kb | 127 SNPs in MR
a
caQTL
2558 variants
eQTL
2603 variants
− log10(P)
chr19 (Mb)
Peak-TSS distance
21.9 kb
chr19:21474531-21475032
ZNF429
b
Mendelian randomization
Causal SMR: p=5.6e-27 | Causal IVW: p=5.4e-53
Causal Egger: p=1.1e-45 | Causal Joint: p=1.6e-52
Causal SMR
Causal IVW
Causal Egger
chr19_21587341
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = -0.334
p=0.002
80 metacells
CD8 Naive
B memory
MAIT
NK
CD4 Naive
CD8 Memory
B naive
B intermediate
Treg
CD4 Memory
CD14 Mono
CD16 Mono
pDC
cDC
RNA log1p(CPM)
ATAC log1p(CPM)
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.384
p=1.3e-13
346 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

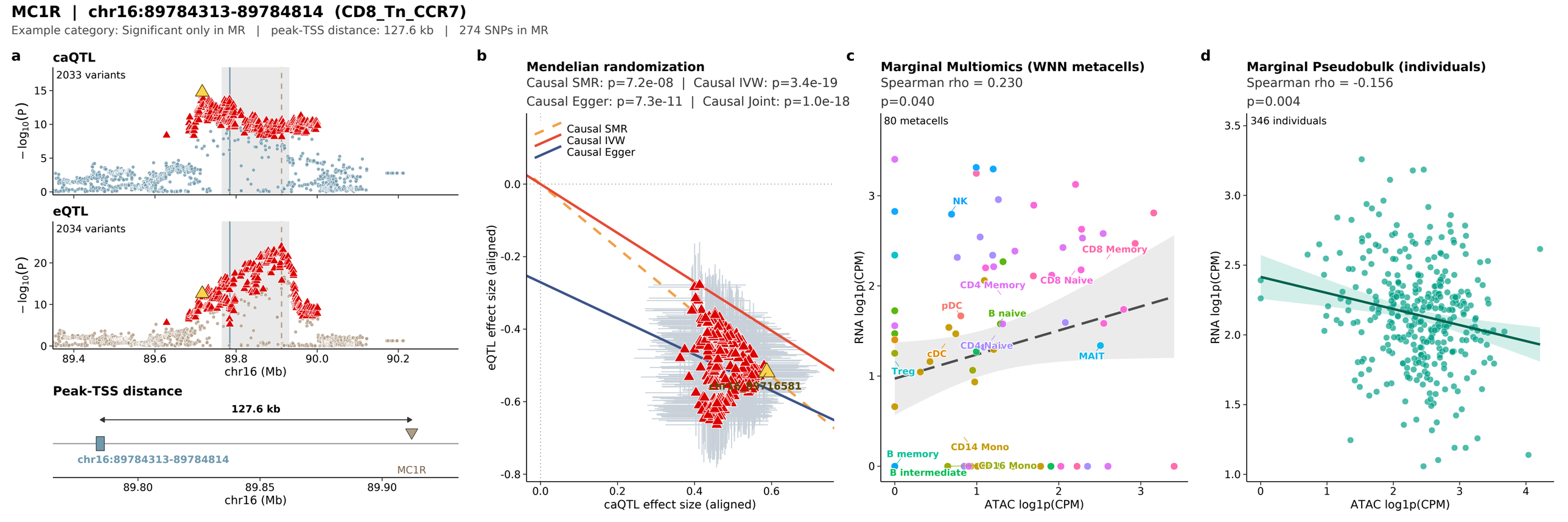

RPS26 | chr12:56041553-56042054 (CD8_Tn_CCR7)
Example category: Opposite direction (MR vs. marginal) | peak-TSS distance: 453 bp | 55 SNPs in MR
a
caQTL
1050 variants
eQTL
1047 variants
− log10(P)
chr12 (Mb)
Peak-TSS distance
453 bp
chr12:56041553-56042054
RPS26
b
Mendelian randomization
Causal SMR: p=1.4e-52 | Causal IVW: p=4.9e-42
Causal Egger: p=7.7e-27 | Causal Joint: p=4.1e-52
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
chr12_56034021
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.748
p=1.5e-15
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
CD4 Memory
Treg
MAIT
CD4 Naive
B memory
CD8 Naive
CD8 Memory
B intermediate
NK
B naive
cDC
CD16 Mono
pDC
CD14 Mono
d
Marginal Pseudobulk (individuals)
Spearman rho = -0.614
p=2.9e-37
346 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

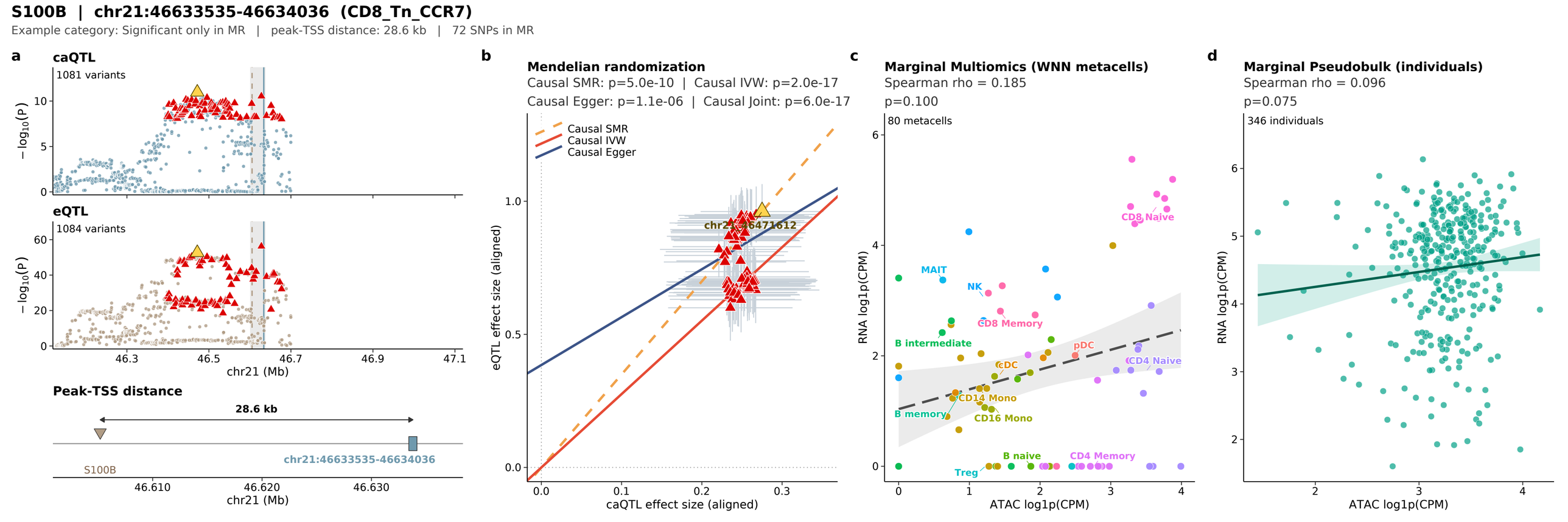
ERAP2 | chr5:96896259-96896760 (CD8_Tn_CCR7)

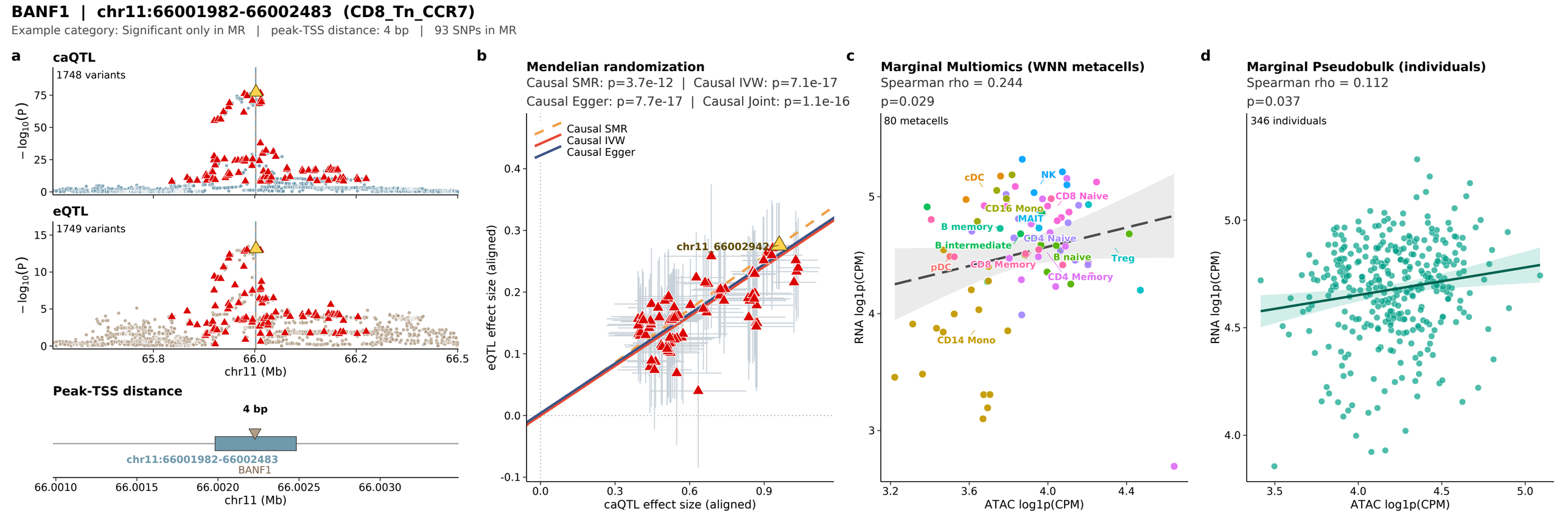
Example category: Opposite direction (MR vs. marginal) | peak-TSS distance: 20.6 kb | 58 SNPs in MR

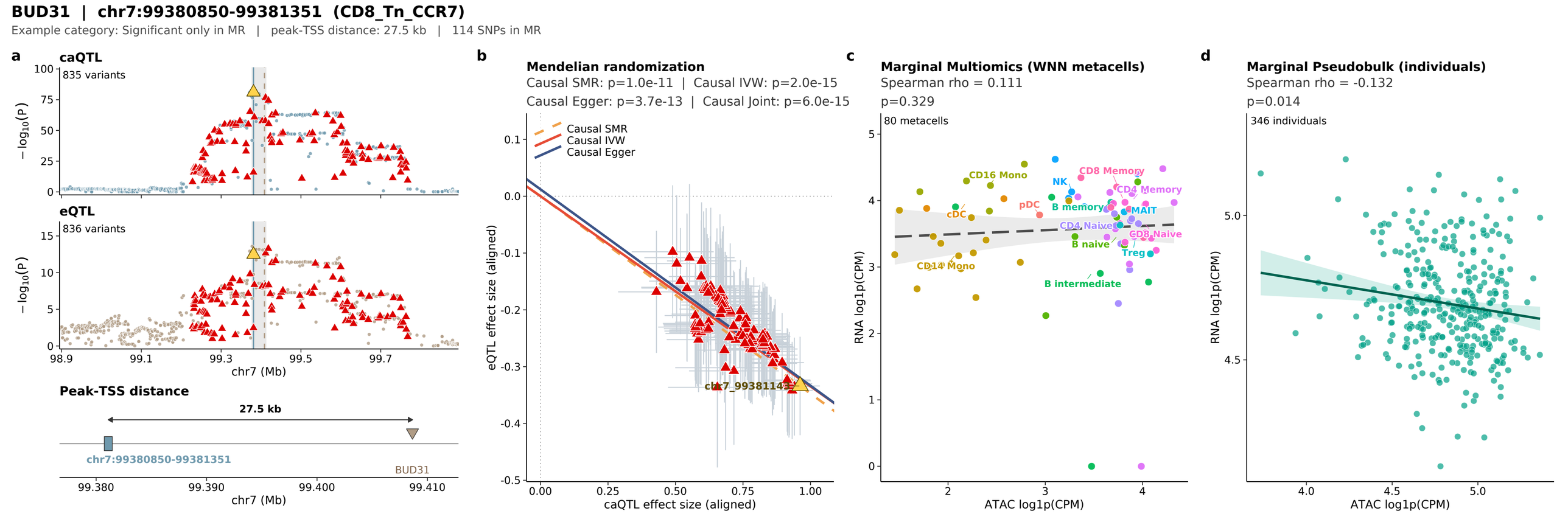


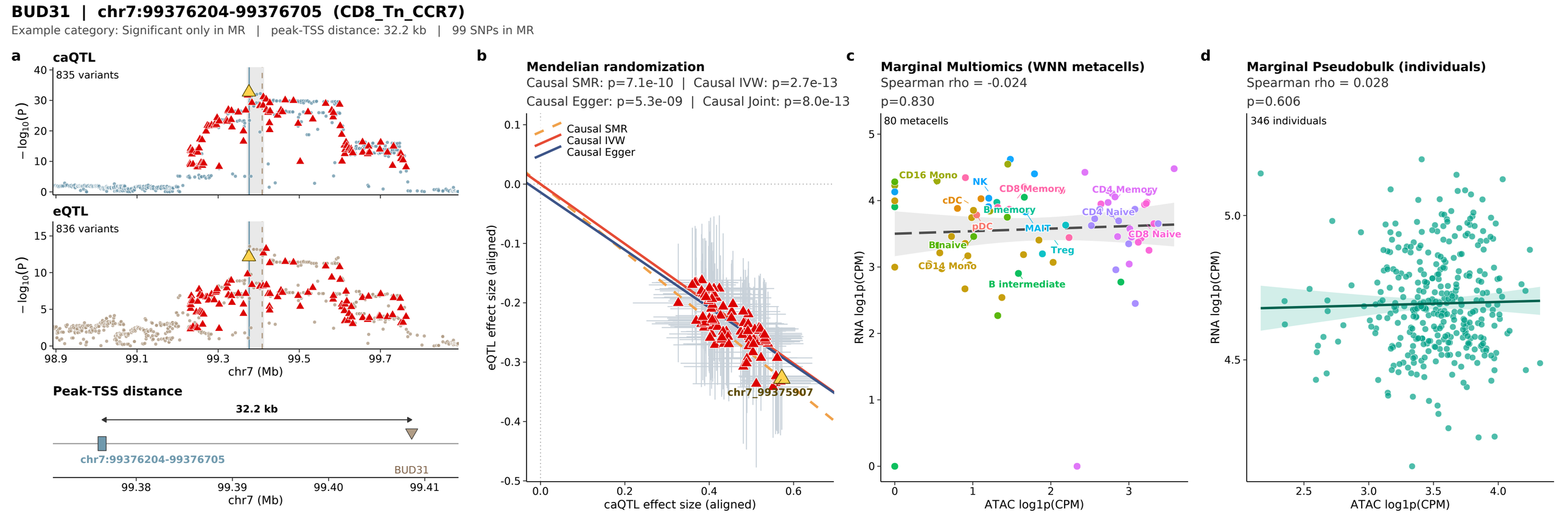

ARPC1B | chr7:99376204-99376705 (CD8_Tn_CCR7)
Example category: Significant only in MR | peak-TSS distance: 2.2 kb | 100 SNPs in MR
a
caQTL
829 variants
− log10(P)
eQTL
830 variants
chr7 (Mb)
Peak-TSS distance
2.2 kb
chr7:99376204-99376705
ARPC1B
b
Mendelian randomization
Causal SMR: p=5.0e-21 | Causal IVW: p=5.1e-39
Causal Egger: p=3.0e-24 | Causal Joint: p=1.5e-38
Causal SMR
Causal IVW
Causal Egger
chr7_99375901
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = -0.149
p=0.186
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
cDC
CD16 Mono
NK
MAIT
CD8 Memory
CD4 Memory
Treg
B memory
B intermediate
pDC
CD8 Naive
CD14 Mono
CD4 Naive
B naive
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.095
p=0.076
346 individuals
RNA log1p(CPM)
ATAC log1p(CPM)


MC1R | chr16:89784313-89784814 (CD8_Tn_CCR7)
Example category: Significant only in MR | peak-TSS distance: 127.6 kb | 274 SNPs in MR
a
caQTL
2033 variants
eQTL
2034 variants
Peak-TSS distance
127.6 kb
chr16:89784313-89784814
MC1R
chr16 (Mb)
b
Mendelian randomization
Causal SMR: p=7.2e-08 | Causal IVW: p=3.4e-19
Causal Egger: p=7.3e-11 | Causal Joint: p=1.0e-18
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.230
p=0.040
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
NK
CD8 Memory
CD4 Memory
CD8 Naive
pDC
B naive
cDC
CD4 Naive
MAIT
Treg
B memory
CD14 Mono
B intermediate
CD16 Mono
d
Marginal Pseudobulk (individuals)
Spearman rho = -0.156
p=0.004
346 individuals
RNA log1p(CPM)
ATAC log1p(CPM)


S100B | chr21:46633535-46634036 (CD8_Tn_CCR7)
Example category: Significant only in MR | peak-TSS distance: 28.6 kb | 72 SNPs in MR
a
caQTL
1081 variants
− log10(P)
eQTL
1084 variants
chr21 (Mb)
Peak-TSS distance
28.6 kb
chr21:46633535-46634036
S100B
b
Mendelian randomization
Causal SMR: p=5.0e-10 | Causal IVW: p=2.0e-17
Causal Egger: p=1.1e-06 | Causal Joint: p=6.0e-17
Causal SMR
Causal IVW
Causal Egger
chr21:46471612
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.185
p=0.100
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
CD8 Naive
MAIT
NK
CD8 Memory
B intermediate
pDC
cDC
CD4 Naive
CD14 Mono
B memory
CD16 Mono
B naive
CD4 Memory
Treg
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.096
p=0.075
346 individuals
RNA log1p(CPM)
ATAC log1p(CPM)


BANF1 | chr11:66001982-66002483 (CD8_Tn_CCR7)
Example category: Significant only in MR | peak-TSS distance: 4 bp | 93 SNPs in MR
a
caQTL
1748 variants
eQTL
1749 variants
− log10(P)
chr11 (Mb)
Peak-TSS distance
4 bp
chr11:66001982-66002483
BANF1
b
Mendelian randomization
Causal SMR: p=3.7e-12 | Causal IVW: p=7.1e-17
Causal Egger: p=7.7e-17 | Causal Joint: p=1.1e-16
Causal SMR
Causal IVW
Causal Egger
chr11_66002942
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.244
p=0.029
80 metacells
cDC
NK
CD8 Naive
CD16 Mono
MAIT
B memory
CD4 Naive
B intermediate
B naive
Treg
pDC
CD8 Memory
CD4 Memory
CD14 Mono
RNA log1p(CPM)
ATAC log1p(CPM)
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.112
p=0.037
346 individuals
RNA log1p(CPM)
ATAC log1p(CPM)


BUD31 | chr7:99380850-99381351 (CD8_Tn_CCR7)
Example category: Significant only in MR | peak-TSS distance: 27.5 kb | 114 SNPs in MR
a
caQTL
835 variants
eQTL
836 variants
− log10(P)
chr7 (Mb)
Peak-TSS distance
27.5 kb
chr7:99380850-99381351
BUD31
b
Mendelian randomization
Causal SMR: p=1.0e-11 | Causal IVW: p=2.0e-15
Causal Egger: p=3.7e-13 | Causal Joint: p=6.0e-15
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.111
p=0.329
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
CD16 Mono
CD8 Memory
NK
CD4 Memory
pDC
B memory
cDC
MAIT
CD4 Naive
CD8 Naive
B naive
Treg
CD14 Mono
B intermediate
d
Marginal Pseudobulk (individuals)
Spearman rho = -0.132
p=0.014
346 individuals
RNA log1p(CPM)
ATAC log1p(CPM)


BUD31 | chr7:99376204-99376705 (CD8_Tn_CCR7)
Example category: Significant only in MR | peak-TSS distance: 32.2 kb | 99 SNPs in MR
a
caQTL
835 variants
eQTL
836 variants
− log10(P)
chr7 (Mb)
Peak-TSS distance
32.2 kb
chr7:99376204-99376705
BUD31
b
Mendelian randomization
Causal SMR: p=7.1e-10 | Causal IVW: p=2.7e-13
Causal Egger: p=5.3e-09 | Causal Joint: p=8.0e-13
Causal SMR
Causal IVW
Causal Egger
chr7_99375907
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = -0.024
p=0.830
80 metacells
CD16 Mono
NK
CD8 Memory
CD4 Memory
cDC
B memory
CD4 Naive
pDC
MAIT
CD8 Naive
B naive
Treg
CD14 Mono
B intermediate
RNA log1p(CPM)
ATAC log1p(CPM)
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.028
p=0.606
346 individuals
RNA log1p(CPM)
ATAC log1p(CPM)